# Classical Mechanics - Home Assignment 5

**Hamilton-Jacobi Equation and Action-angle Variable (November 2024)**

1. For planetary orbits in three dimensions, the Hamiltonian is given by

$$h = \frac{1}{2m}\left[p_r^2 + \frac{1}{r^2}p_\theta^2 + \frac{1}{r^2\sin^2\theta}p_\phi^2\right] - \frac{k}{r}$$

   Find Hamilton's principal function in trems of integrals over $r$ and $\theta$.

2. A particle slides down a cycloidal track

$$x = l(2\phi + \sin 2\phi), \qquad y = l(1 - \cos 2\phi)$$

   in a vertical plane without friction under gravity. Using action-angle variables, find the frequency of oscillation for $\phi \leq \pi/4$.

3. A particle of charge $q$ is moving in a circle in X-Y plane due to a magnetic field $\vec{B}$ along Z axis. Find the adiabatic invariant when the field varies adiabatically along Z.

4. For a pendulum whose amplitude is not necessarily small, derive the action integral and hence the time period. (The result will be in the form of an integral.)